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2.5 Cytoplasm and Cytoskeleton

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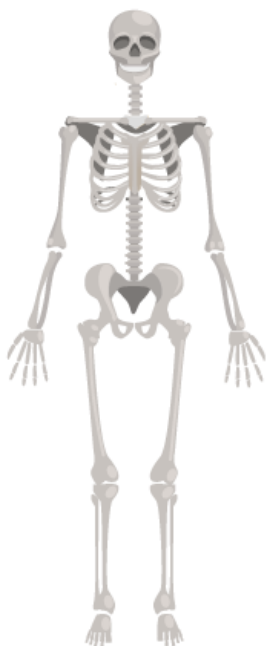
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Fact-checked by: The CK-12 Editorial Team

Last Modified: May 01, 2026

What you will learn

- Structure and function of the **cytoplasm**
- Structure and function of the **cytoskeleton**
- Three main types of cytoskeleton fibers



[Figure 1]

Does a cell have, or even need, a "skeleton"?



basis of the "skeleton" inside the **cell**.

The Cytoplasm and Cytoskeleton

The **cytoplasm** consists of everything inside the **cell membrane** of the cell, excluding the **nucleus** in a **eukaryotic cell**. It includes the watery, gel-like material called **cytosol**, as well as various structures. The water in the cytoplasm makes up about two-thirds of the cell's weight and gives the cell many of its properties.

Functions of the Cytoplasm

The cytoplasm has several important functions, including:

1. suspending cell **organelles**.
2. pushing against the plasma membrane to help the cell keep its shape.
3. providing a site for many of the **biochemical reactions** of the cell.

Cytoplasm-The important fluid of the cell

MooMooMath and Science



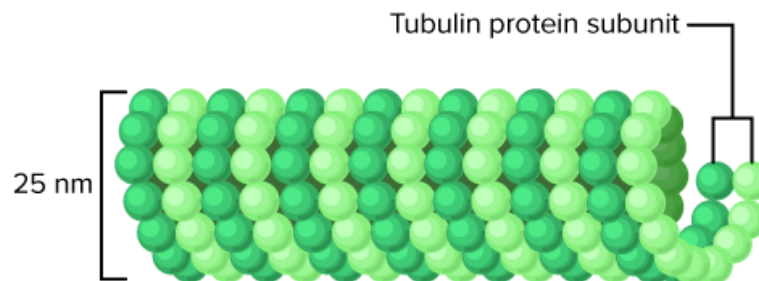
Watch on

The Cytoskeleton

The **cytoskeleton** is a cellular "scaffolding" or "skeleton" that crisscrosses the cytoplasm. All eukaryotic cells have a cytoskeleton, and recent research has shown that **prokaryotic cells** also have a cytoskeleton. The eukaryotic cytoskeleton is made up of a network of long, thin **protein** fibers and has many functions. It helps to maintain cell shape. It holds organelles in place, and for some cells, it enables cell **movement**. The cytoskeleton also plays important roles in both the intracellular movement of substances and in **cell division**. Certain proteins act like a path that **vesicles** and organelles move along within the cell. The threadlike proteins that make up the cytoskeleton continually rebuild to adapt to the cell's constantly changing needs.

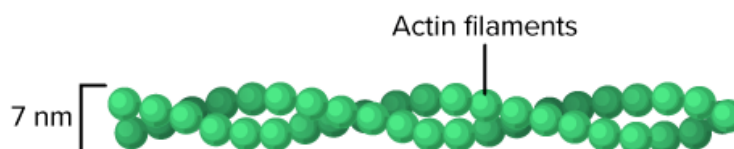
The main kinds of cytoskeleton fibers are microtubules, **intermediate filaments**, and microfilaments.

forms microtubules. Two forms of tubulin, alpha and beta, form dimers (pairs) which come together to form the hollow cylinders. The cylinders are twisted around each other to form the microtubules. Microtubules help the cell keep its shape as shown in the **Figure below**. They hold organelles in place and allow them to move around the cell, and they form the mitotic **spindle** during cell division. Microtubules also make up parts of **cilia** and **flagella**, the organelles that help a cell move.



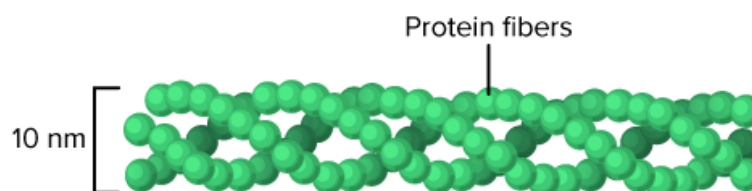
[Figure 2]

- **Microfilaments**, shown in the **Figure below**, are made of two thin **actin** chains that are twisted around one another. Microfilaments are mostly concentrated just beneath the cell membrane, where they support the cell and help the cell keep its shape. Microfilaments form cytoplasmic extensions, such as pseudopodia and microvilli, which allow certain cells to move. The actin of the microfilaments interacts with the protein **myosin** to cause contraction in muscle cells. Microfilaments, as shown in the **Figure below**, are found in almost every cell, and are numerous in muscle cells and in cells that move by changing shape, such as **phagocytes** (white blood cells that search the body for **bacteria** and other invaders).

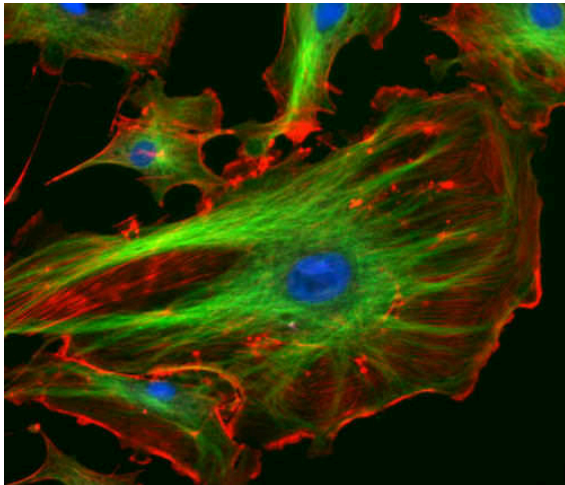


[Figure 3]

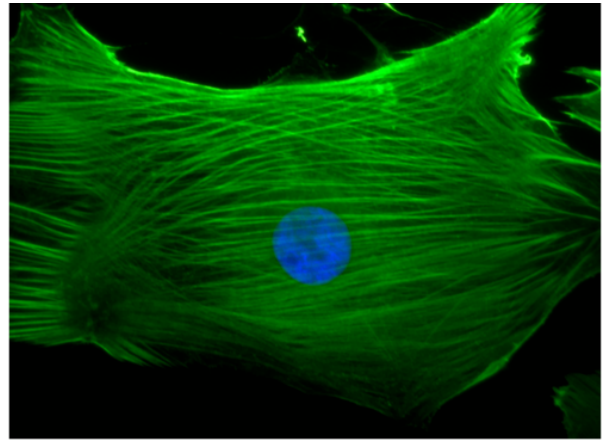
- **Intermediate filaments**, shown in the **Figure below**, differ in make-up from one cell type to another. Intermediate filaments organize the inside structure of the cell by holding organelles and providing strength. They are also structural components of the **nuclear envelope**. Intermediate filaments made of the protein **keratin** are found in skin, hair, and nails cells.



[Figure 4]



[Figure 5]



[Figure 6]

The eukaryotic cytoskeleton.

(Left) Microfilaments are shown in red, microtubules in green, and the nuclei are in blue. By linking regions of the cell together, the cytoskeleton helps support the shape of the cell.

(Right) Microscopy of microfilaments (actin filaments), shown in green, inside cells. The nucleus is shown in blue.

Cytoskeleton Structure

	Microtubules	Intermediate Filaments	Microfilaments
Fiber Diameter	About 25 nm	8 to 11 nm	Around 7 nm
Protein Composition	Tubulin, with two subunits, alpha and beta tubulin	One of different types of proteins such as lamin, vimentin, and keratin	Actin
Shape	Hollow cylinders made of two protein chains twisted around each other	Protein fiber coils twisted into each other	Two actin chains twisted around one another
Main Functions	Organelle and vesicle movement; form mitotic spindles during cell reproduction; cell motility (in cilia and flagella)	Organize cell shape; positions organelles in cytoplasm; structural support of the nuclear envelope and sarcomeres; involved in cell-to-cell and cell-to-matrix junctions	Keep cellular shape; allows movement of certain cells by forming cytoplasmic extensions or contraction of actin fibers; involved in some cell-to-cell or cell-to-matrix junctions





DID YOU KNOW?

Colchicine is an alkaloid isolated from the plant autumn crocus. This alkaloid prevents the assembly of the microtubules and does not allow the formation of spindle fibers during cell division. This stops the cell in the **metaphase** stage and blocks cell division. Since colchicine does not stop **DNA replication** it allows doubling of **chromosomes** when applied to meristematic **tissues** in plants.

Check your understanding!

Summary

- The cytoplasm consists of everything inside the cell membrane of the cell, excluding the nucleus in a eukaryotic cell.
- The cytoskeleton is a cellular "skeleton" that crisscrosses the cytoplasm. The three main cytoskeleton fibers are microtubules, intermediate filaments, and microfilaments.
- Microtubules are the thickest of the cytoskeleton structures and are most commonly made of filaments which are polymers of alpha and beta tubulin.
- Intermediate filaments organize the inside structure of the cell by holding organelles and providing strength.

Microfilaments are the thinnest of the cytoskeleton structures and are made of two thin actin chains that are twisted around one another.

- 1. What is the difference between cytoplasm and cytosol?
- 2. List two roles of the cytoplasm.
- 3. Name the three main types of cytoskeleton fibers.
- 4. List two functions of the eukaryotic cytoskeleton.

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